

LE NGUYEN DUY LUAN

Mobile Developer | On-device AI / Computer Vision
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PROFESSIONAL SUMMARY

Motivated and detail-oriented Mobile Developer specializing in Native Android (Kotlin, Jetpack Compose) and On-device AI/Computer Vision (TensorFlow Lite). Passionate about building performance-optimized, offline-first mobile applications with real-time ML capabilities. Possesses strong problem-solving skills, solid understanding of fullstack architectures, and a dedication to writing clean, maintainable code. Actively seeking opportunities to contribute to innovative software products.

EDUCATION

Nha Trang University

Bachelor of Information Technology

Mar 2026

Nha Trang, Vietnam

- **GPA:** 7.94/10 (3.25/4.0)
- **Graduation Project:** Lung X-ray Disease Classification using Segmentation-guided Masking — Score: 9.1/10
- **Relevant Coursework:** Mobile Application Development, Web Application Development, Image Processing, Database Management, Artificial Intelligence

TECHNICAL SKILLS

Languages: Java, Kotlin, JavaScript, Python, SQL

Mobile: Android SDK, CameraX, Jetpack components, Room (local database)

AI / Computer Vision: TensorFlow Lite, YOLO (object detection), OpenCV, model conversion & on-device deployment

Web / Backend: REST API design, Node.js basics, fullstack frameworks, online database integration

Tools: Git/GitHub, Android Studio, VS Code, Figma (basic)

Languages (Spoken): Vietnamese (native), English (technical reading/writing)

PROJECTS

VocabLensAI — Native Android English Learning Platform | *Personal Project*

- **Tech Stack:** Kotlin, Jetpack Compose, Coroutines/Flow, Room DB, CameraX, TensorFlow Lite, Gemini API, Firebase (Auth & Firestore), Supabase.
- Architected and developed an offline-first Android application from scratch using Kotlin, Jetpack Compose, MVVM, and clean code practices.
- Implemented on-device real-time object detection by integrating CameraX and a lightweight TensorFlow Lite (.tflite) model to instantly extract and generate vocabulary flashcards from real-world surroundings.
- Integrated Gemini AI to build a contextual story generator (AI Storyteller) and a phonetic pronunciation analyzer, incorporating local caching to respect API token rate limits.
- Designed a custom learning engine incorporating the SM-2 Spaced Repetition algorithm (calculating Ease Factor, repetitions, and intervals) to automatically schedule daily reviews.
- Built a real-time multiplayer PvP Arena using Firebase Firestore, implementing a matchmaking system with simulated bot fallbacks and an ELO rating calculator ($K = 40/20$).
- Developed Chess.com-style game review UI displaying match accuracy (%), Genius/Blunder move tags, and dynamically generated AI feedback on gameplay performance.
- Designed a hybrid data sync system utilizing Room Database for local caching and offline operations, paired with Firestore for background profile and progress synchronization.

Lung X-ray Disease Classification | *Graduation Project, Nha Trang University*

- **Tech Stack:** Python, TensorFlow/Keras, OpenCV, U-Net (Segmentation), ResNet/DenseNet (Classification).
- Developed a segmentation-guided masking pipeline to isolate lung regions in chest X-ray images before classification, improving model focus on diagnostically relevant areas and reducing noise.
- Trained, validated, and fine-tuned deep learning classification models, achieving a high degree of technical accuracy and a final project score of 9.1/10.
- Documented the end-to-end workflow from data preprocessing and image augmentation to model evaluation, creating a foundation suitable for medical-imaging research handoffs.

Fullstack Task & Inventory Management System | *Practice Project*

- **Tech Stack:** React.js, Node.js, Express, MongoDB, RESTful APIs, JWT Authentication.
- Developed a robust fullstack web application featuring secure user authentication, role-based access control, and dynamic inventory tracking.
- Architected a scalable backend with an online database, replacing an earlier hardcoded REST API approach to simulate a production-realistic environment.
- Implemented responsive frontend interfaces for seamless data entry, querying, and dashboard analytics.